

Contestant Number: _____

Time: _____

Rank: _____



NETWORK ADMINISTRATION USING CISCO (315) STATE 2021

TOTAL POINTS

_____ (100 points)

Test Time: 60 minutes

Please select the answer that best answers the question.

1. What command is used to encrypt all passwords on the Cisco switch?
 - a. service password-encryption
 - b. enable secret level 15 0 <password>
 - c. encrypt password all
 - d. keychain encrypt all
2. What command is used to clear all mac address table entries?
 - a. clear mac-address-table
 - b. erase mac-address-table
 - c. delete mac-address-table
 - d. remove mac-address-table
3. What command enables an instance of OSPF on a Cisco router?
 - a. router ospf
 - b. ip routing
 - c. ip router opsf 1
 - d. ip route open-shortest-path-first
4. What command is used to display the current status of Spanning-Tree and how it is affecting ports on the Cisco Switch?
 - a. show spanning-tree
 - b. show spanning-tree online
 - c. show port-forwarding
 - d. show pvlan-spanning-tree
5. What command is used to configure the Cisco switch to use rapid per-vlan spanning-tree protocol?
 - a. enable rapid-pvst
 - b. set spanning-tree mode rapid-pvst
 - c. spanning-tree rapid-pvst enable
 - d. spanning-tree mode rapid-pvst
6. What subnet mask could be applied to the network 192.168.10.0 which would most closely allow 29 hosts on the subnetwork?
 - a. 255.255.255.0
 - b. 255.255.255.192
 - c. 255.255.255.240
 - d. 255.255.255.224

7. What subnet mask could be applied to the network 192.168.25.0 which would closely create at least 9 subnetworks?
 - a. 255.255.255.224
 - b. 255.255.255.192
 - c. 255.255.255.240
 - d. 255.255.255.128

8. Which IP Addresses properly exists in the network 192.168.52.16/28?
 - a. 192.168.52.29
 - b. 192.168.52.32
 - c. 192.168.52.10
 - d. 192.168.52.33

9. Which IP Address would be able to communicate with the IP Address 192.168.39.147/26?
 - a. 192.168.39.87
 - b. 192.168.39.129
 - c. 192.168.39.202
 - d. 192.168.39.1

10. Which of the following would *not* be a network ID produced when using the IP network 192.168.4.0 and the CIDR mask /27?
 - a. 192.168.4.16
 - b. 192.168.4.32
 - c. 192.168.4.64
 - d. 192.168.4.128

11. When connecting a host to a port on a switch, the port does *not* turn on. Which of the following is the *most* likely cause?
 - a. RIP has disabled the port.
 - b. Spanning-Tree has marked the port as a designated port.
 - c. The cable between the host and the switch is not a straight cable.
 - d. The port on the switch has not been enabled with the command “turn on port”.

12. With a new host that is connected to a switch and given an appropriate IP Address, it is *not* able to reach an existing host that is on the same IP network. What is the most likely reason it *cannot* communicate?
 - a. The port on the switch that the new host is connected to is in another VLAN.
 - b. The default gateway for the new host is set incorrectly.
 - c. The existing host does not have a route back to the new host.
 - d. OSPF is blocking the port to the new host.

13. When configuring a new trunk between two switches, hosts on all VLANs *except* for the Marketing VLAN are able to communicate with their other respective hosts. What is the most likely reason the Marketing VLAN *cannot* communicate between the two switches?
- The “no shut” command has not been entered for the VLAN on both switches.
 - A firewall on the switch is preventing the VLAN from communicating.
 - The trunk port is not in access mode for the Marketing VLAN.
 - The Marketing VLAN was not added to one or both sides of the trunk connection.
14. On a Cisco router, after setting the IP Address on an interface, you are *not* able to ping it. What is most likely the problem?
- Spanning-tree has disabled the port.
 - The IP Address assigned to the port is a network ID.
 - The port has not been enabled with the “no shut” command.
 - RIP has not been enabled on the router.
15. A Cisco router has its interface configured for routing which includes having an IP Address set on it and the port being enabled. However, an entry for the new network is *not* added into the routing table. What is the most likely issue?
- The port is not physically connected.
 - The routing table is full and cannot accept new entries.
 - The port has an IP address on it that is in the same network as another port on the router.
 - The command “enable routing” has not been configured on the port.
16. Which commands below configures a port on a Cisco switch to be a trunk port and allow just VLANs 10 and 20?
- switchport trunk enable; switchport trunk enable vlan 10,20
 - switchport mode trunk; switchport trunk allow vlan 10,20
 - switchport mode trunk; switchport trunk allow all
 - switchport trunk enable; switchport trunk enable all
17. In the sense of Spanning-Tree Protocol, what is the effect of enable “PortFast” on an interface?
- It enables accelerated transfer speeds on the interface, thereby lowering the port cost.
 - It removes the interface from the Spanning-Tree environment.
 - It makes the interface skip the listening and learning states and go straight to the forwarding state.
 - It enables Rapid Spanning-Tree Protocol on the interface.

18. On a Cisco switch, you run the command “show spanning-tree active” and receive the following output:

Interface	Role	Sts	Cost	Prio.	Nbr	Type	
-----	-----	-----	-----	-----	-----	Fa0/13	Altn
BLK 19	128.13	P2p					
Fa0/11	Root	FWD	19	128.11	P2p		

What does the role “Root” most likely mean?

- This port is used to connect this switch to the upstream switch in the direction of the root switch.
 - This port is used by downstream switches to connect to it since it is the root switch.
 - This port is the bottom of the spanning-tree environment.
 - This port is being blocked to prevent a layer-2 loop.
19. On a Cisco switch, you run the command “show spanning-tree active” and receive the following output:

Interface	Role	Sts	Cost	Prio.	Nbr	Type	
-----	-----	-----	-----	-----	-----	Fa0/13	Desg
FWD 19	128.13	P2p					
Fa0/16	Desg	FWD	19	128.16	P2p		

What does the role “Desg” most likely mean?

- This port is used to connect this switch to the upstream switch in the direction of the root switch.
 - The port is being blocked to prevent a layer-2 loop.
 - Port 13 and 16 are connected to one another using a cross-over cable.
 - This port is used by downstream switches to connect to it since it is in the direction of the root switch.
20. What are the states of 802.1D Spanning-Tree?
- Disabled, Blocking, Listening, Learning, Forwarding
 - Discarding, Learning, Forwarding
 - Removing, Accepting, Learning, Forwarding
 - Disabled, Enabled
21. Which of the following is the IEEE standard for Rapid Spanning-Tree?
- 802.1w
 - 802.1d
 - 802.11
 - 802.11a

22. On a Cisco switch, you run the command “show spanning-tree active” and receive the following output:

```
Switch#show span active
VLAN0001
Spanning tree enabled protocol ieee
Root ID      Priority 32769
              Address 0009.7CC4.EBE4
              Cost 19
              Port 11(FastEthernet0/11)
              Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID    Priority 32769 (priority 32768 sys-id-ext 1)
              Address 0010.1148.23B8
              Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
              Aging Time 20
```

What is the significance of “Address 0009.7CC4.EBE4”?

- a. It is the MAC address of this switch.
 - b. This is the MAC address of root switch in the spanning-tree environment.
 - c. It is the MAC address of the next downstream switch.
 - d. It is the MAC address of the host connected to port FastEthernet0/11.
23. What are the states of the 802.1W Rapid Spanning-Tree?
- a. Discarding, Learning, Forwarding
 - b. Disabled, Blocking, Listening, Learning, Forwarding
 - c. Removing, Accepting, Learning, Forwarding
 - d. Disabled, Enabled
24. In the sense of Rapid Spanning-Tree, what does the port role “Alternate” mean?
- a. This port is the backup “root” port, to be used if the current root port fails.
 - b. This port is the backup “designated” port, to be used if the current designated port fails.
 - c. This signifies that this switch will take over as root if the root switch fails.
 - d. This signifies that this switch will take over as the designated switch if the current designated switch fails.
25. In the sense of Rapid Spanning-Tree, what does the port role “Backup” mean?
- a. This port is the backup “root” port, to be used if the current root port fails.
 - b. This signifies that this switch will take over as root if the root switch fails.
 - c. This port is the backup “designated” port, to be used if the current designated port fails.
 - d. This signifies that this switch will take over as the designated switch if the current designated switch fails.

26. Router Information Protocol (RIP) is a Distance Vector Routing Protocol.
- True
 - False
27. What is the maximum number of hops allowed in a routing environment using RIP?
- 8
 - 15
 - 16
 - 32
28. What is the correct command to enable RIP on an interface which has an IP Address of 192.168.37.1?
- network 192.168.37.1 255.255.255.0
 - network 192.168.37.0 0.0.0.255
 - network 192.168.37.1/24
 - network 192.168.37.0
29. When looking at the output of “show ip route”, what code indicates a route learned by RIP?
- S
 - C
 - R
 - O
30. Open Shortest Path First (OSPF) is a Distance Vector Routing Protocol.
- True
 - False
31. What information is shared within a single area of OSPF?
- Link-states within the databases of all routers in the area
 - Network summaries of all networks for the routers in the area
 - Router management IP Addresses for routers in the area
 - Distance between the router interfaces
32. What is the correct command to enable single area OSPF on an interface which has an IP Address of 192.168.37.1/24?
- network 192.168.37.0 0.0.0.255 area 0
 - network 192.168.37.0 area 0
 - network 192.168.37.0 255.255.255.0 area 0
 - network 192.168.37.0 0.0.0.255

33. When turning on OSPF for a router using the command “router ospf 1”, what does the “1” signify?
- The Process ID for OSPF on this router.
 - The autonomous system number which should match all other routers in the area.
 - The priority for being the designated router.
 - This will be the first router with OSPF enabled.
34. When a router has learned the same route using RIP and OSPF, OSPF will take precedence.
- True
 - False
35. With OSPF, if the Router-ID has *not* manually been set, what is the next source it will use for its Router-ID?
- Highest IP Address assigned on the device
 - IP Address of Loopback Interface
 - Lowest IP Address assigned on the device
 - IP Address of Management Interface
36. What range of values are used to create a standard access control list?
- 100-199
 - 1-499
 - 1-99
 - 500-999
37. Standard access control list can be used to identify traffic based on destination information.
- True
 - False
38. Which option is commonly used to identify a single IP Address in an access control list?
- any
 - subnet mask
 - host
 - wildcard mask
39. What is the effect of the following access control list entries (assuming they have the same ACL number)?
- ```
deny ip host 192.168.52.70
permit ip 192.168.52.0 0.0.0.255
```
- It will block address 192.168.52.70 but allow all other traffic.
  - It will block all 192.168.52.0/24 traffic.
  - It will block all traffic except for 192.168.52.70.
  - It will allow all traffic.

40. When placing an ACL on an interface, what direction is the ACL allowed to check?
- a. In or out
  - b. In
  - c. Out
  - d. Away
41. There is a requirement that all management communication made to the Cisco device must be secure. What protocol should be enabled and used?
- a. Telnet
  - b. HTTP
  - c. FTP
  - d. SSH
42. What port number is used by SSH?
- a. 21
  - b. 22
  - c. 80
  - d. 443
43. What port number is used by Telnet?
- a. 21
  - b. 22
  - c. 23
  - d. 80
44. What port is commonly used for secure HTTP traffic?
- a. 22
  - b. 80
  - c. 443
  - d. 8080
45. FTP send data between the client and server securely.
- a. True
  - b. False
46. What is an advantage of using fiber optic cables instead of copper?
- a. Communication can occur much greater distances.
  - b. It's a common media type used by PCs and laptops.
  - c. It is easier to assemble connectors on cable than on copper cables.
  - d. They are more durable than copper cables.

47. Based on the following starting network and subnet mask, which of the following addresses is *not* assignable to a device?

192.168.13.0 255.255.255.224

- a. 192.168.13.64
- b. 192.168.13.54
- c. 192.168.13.34
- d. 192.168.13.24

48. Which command enables IPv6 forwarding on a Cisco router?

- a. ipv6 routing enable
- b. ipv6 enable
- c. no ipv6 disable
- d. ipv6 unicast-routing

49. What command can be used to identify what VLANs are allowed on a trunk port?

- a. show vlan
- b. show interface
- c. show interface switchport
- d. show interface vlan

50. What is the keyboard shortcut to return to privilege user mode directly?

- a. Ctrl+Z
- b. Ctrl+C
- c. Ctrl+B
- d. Ctrl+X